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coach

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1. What is the role of a loss function in training a neural network?

1 / 1 point

- ☒ To measure the difference between predicted and actual outcomes
- ☐ To split the dataset into training and testing sets
- ☐ To initialize the neural network parameters
- ☐ To update the weights of the network

 **Correct**
Correct! The role of a loss function is to measure the difference between the predicted and actual outcomes, guiding the learning process.

2. What is the first step in the process of building a model for multi-class classification using PyTorch?

1 / 1 point

- ☒ Data Setup
- ☐ Model Evaluation
- ☐ Model Training
- ☐ Hyperparameter Tuning

 **Correct**
Correct! Data setup is the first step in building a model for multi-class classification. It involves loading and preparing the dataset for training.

3. In a Convolutional Neural Network (CNN), what is the purpose of using convolutional filters?

1 / 1 point

- ☒ To detect specific features such as edges or textures in the input image.
- ☐ To prevent overfitting during the training process.
- ☐ To normalize the input data.
- ☐ To reduce the number of parameters in the network.

 **Correct**
Correct! Convolutional filters are used to detect specific features like edges and textures which are critical for image recognition.

4. What are some common methods of data augmentation used in training Convolutional Neural Networks (CNNs)?

0.8 / 1 point

☒ Flipping

 **Correct**
Correct! Flipping, particularly horizontal flipping, is widely used in data augmentation.

☒ Color Shifting

 **This should not be selected**
Incorrect. While color shifting can be used, it is less common compared to other techniques like rotation and flipping.

☐ Image Denoising

☒ Zooming

 **Correct**
Correct! Zooming in and out is a popular data augmentation method that helps the model generalize better.

☒ Rotation

 **Correct**
Correct! Rotation is a common data augmentation technique used to make the model invariant to image rotations.

5. Which package is commonly used to create spectrograms in Python?

1 / 1 point

- ☐ seaborn
- ☐ pandas
- ☐ numpy
- ☒ librosa

 **Correct**
Correct! 'librosa' is a popular package for music and audio analysis, including the creation of spectrograms.